Quantum Benchmarks

Hardware Runs

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This benchmarking report was generated using Classiq v1.6.0.
For further information, see the Benchmarking directory in classiq-library.

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Adder

We benchmark an out-of-place addition by a constant,

$$|x\rangle_n|0\rangle_n \to |x\rangle_n|x + c \bmod 2^n\rangle_n,$$

where c is a fixed integer. In the benchmark, the input variable is prepared in a uniform superposition, and the constant is chosen as

$$c = 2^{\lfloor n/2 \rfloor} - 1.$$

The score is defined as the total measured probability of obtaining the correct arithmetic relation,

$$\text{Score} = \sum_{x,y=0}^{2^n-1} P(x,y)\, \delta_{\,y,\ x+c \bmod 2^n},$$

where $P(x, y)$ is the measured joint output distribution of the two variables. Equivalently, the score is the total probability mass on outcomes satisfying

$$y = x + c \bmod 2^n.$$

A score of 1 corresponds to perfect agreement with the ideal adder transformation.

adder - 4 qubits (1000 shots)

Provider	Backend Name	Score	Time Elapsed (min)	Depth	2Q Gate Count
Classiq	simulator	1.0	0.0	104	92
Classiq	simulator_statevector	1.0	0.0	104	92
IBM Quantum	ibm_kingston	0.204	395.2	366	170
IonQ	qpu.forte-1	0.877	15.1	108	92

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GHZ State Preparation

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ghz - 3 qubits (100 shots)

Provider	Backend Name	Score	Time Elapsed (min)	Depth	2Q Gate Count
Classiq	simulator	1.0	0.0	4	2
Classiq	simulator_statevector	1.0	0.0	4	2
IBM Quantum	ibm_kingston	0.4768	379.9	12	2
IonQ	qpu.forte-1	0.9609	10.3	8	2

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Grover Search Algorithm

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grover - 4 qubits (1000 shots)

Provider	Backend Name	Score	Time Elapsed (min)	Depth	2Q Gate Count
Classiq	simulator	0.9984	0.0	175	100
Classiq	simulator_statevector	1.0	0.0	99	64
IBM Quantum	ibm_kingston	0.3082	50.9	488	168
IonQ	qpu.forte-1	0.6241	10.3	201	100

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Dynamical Localization

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localization - 3 qubits (1000 shots)

Provider	Backend Name	Score	Time Elapsed (min)	Depth	2Q Gate Count
Classiq	simulator	0.9154	0.0	–	–
Classiq	simulator_statevector	0.9147	0.0	49	30

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QFT

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qft - 4 qubits (1000 shots)

Provider	Backend Name	Score	Time Elapsed (min)	Depth	2Q Gate Count
Classiq	simulator	1.0	0.0	23	18
Classiq	simulator_statevector	1.0	0.0	23	18
IBM Quantum	ibm_kingston	0.6853	86.6	51	18
IonQ	qpu.forte-1	0.9973	15.1	28	18

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Linear State Preparation

We benchmark the preparation of an n-qubit state with linearly increasing amplitudes,

$$|\psi\rangle = \frac{1}{\sqrt{M}} \sum_{x=0}^{2^n-1} x\,|x\rangle,$$

where

$$M = \sum_{x=0}^{2^n-1} x^2.$$

The score is defined from the total variation distance between the observed output distribution p and the ideal distribution u,

$$D_{\mathrm{TV}}(p,u) = \frac{1}{2} \sum_{i=0}^{2^n-1} |p_i - u_i|.$$

We then define

$$S = 1 - \frac{D_{\mathrm{TV}}(p,u)}{1-1/n}.$$

state_preparation - 4 qubits (1000 shots)

Provider	Backend Name	Score	Time Elapsed (min)	Depth	2Q Gate Count
Classiq	simulator	0.9434	0.0	50	28
Classiq	simulator_statevector	1.0	0.0	50	28
IBM Quantum	ibm_kingston	0.8535	479.3	173	54
IonQ	qpu.forte-1	0.8666	10.1	63	28

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Quantum Volume

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2–4 qubits (10 trials, 1000 shots)

Provider	Backend	Quantum Volume	Trials	Average Elapsed Time (min)
Classiq	simulator	16	30/30	0.01
Classiq	simulator_statevector	16	30/30	0.01
IBM Quantum	ibm_kingston	16	29/30	439.19
IonQ	qpu.forte-1	16	29/30	12.8

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